# RoadSOS

Technical Documentation

Architecture, Algorithms, and Implementation Reference

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**Scope.** This document is the engineering reference for RoadSOS. It covers the request lifecycle, the OpenStreetMap Overpass query layer, the Google Places fallback strategy, the four-tier offline cache hierarchy, the AI triage model, internationalization across 48 languages, the real-time map renderer, crash detection signals, and the deployment topology. Every number cited is grounded in source. Every external dependency is named. No marketing.

## Contents

# 1 System Overview

RoadSOS is a Progressive Web Application that locates emergency-services contacts (hospitals, police, ambulances, fire stations, towing, repair, tyre/puncture, showrooms) within driving distance of a user's GPS position and dispatches their coordinates to pre-configured emergency contacts via WhatsApp or SMS.

## 1.1 Top-Level Topology

```
+-----------------------------------------------------------------+
|                   BROWSER  (PWA, installable)                   |
|                                                                 |
|   React 18  +  Vite 7.3.3  +  Workbox 7 (Service Worker)            |
|   |- Leaflet 1.9.4    (real OSM map, dark tiles)                |
|   |- i18next 26       (48 languages, RTL support)               |
|   |- localStorage     (24h search-result cache)                 |
|   `- bundled JSON     (938 entries x 200 countries)            |
+-----------------------------+-----------------------------------+
                              |  HTTPS  (CORS-allowlisted)
                              v
+-----------------------------------------------------------------+
|              FASTAPI BACKEND  (Render, Python 3.11.9)           |
|                                                                 |
|   uvicorn + asyncio + httpx                                     |
|   |- /search    (4-phase orchestrator)                          |
|   |- /triage    (Gemini 2.5 Flash + rule-based fallback)        |
|   |- /dispatch  (telemetry sink for SOS events)                 |
|   |- /health    (liveness + version)                            |
|   `- /offline-pack  (static facility bundle for warm cache)     |
+----+--------------------+--------------------+------------------+
     |                    |                    |
     v                    v                    v
+-----------+       +-------------+      +---------------+
|   OSM     |       |   Google    |      |  Nominatim    |
| Overpass  |       |   Places    |      | (geocoder)    |
|  (free)   |       |   (paid)    |      |   (free)      |
+-----------+       +-------------+      +---------------+
```

## 1.2 Tech Stack at a Glance

| Layer | Components |
|---|---|
| Frontend runtime | React 18.3.1, React DOM 18.3.1, React Router 7.15.0 |
| Build tool | Vite 7.3.3, vite-plugin-pwa 1.3.0 (`injectManifest`) |
| Service worker | Workbox 7.4.1 (precaching, expiration, routing, strategies) |
| Map | Leaflet 1.9.4 + react-leaflet 4.2.1; CartoDB Dark Matter tiles |
| i18n | i18next 26.2.0 + react-i18next 17.0.8; 48 locales |
| Icons | lucide-react 1.16.0 |
| Backend runtime | FastAPI 0.115.0 + uvicorn 0.30.6, Python 3.11.9 |
| Async HTTP | httpx 0.27.2 |
| AI | Google Gemini 2.5 Flash via REST (no SDK; uses httpx) |
| Phone normalization | phonenumbers 8.13.46 |
| Storage | SQLite via aiosqlite 0.20.0 (offline-pack only) |
| Validation | pydantic 2.9.2 |
| Lint / format / test | ruff 0.11.13, pytest 8.3.3, pytest-asyncio 0.24.0, vitest 2.1.4 |
| Deployment | Render (backend), Vercel (frontend) |
| CI/CD | GitHub Actions: `frontend-ci.yml`, `backend-tests.yml`, `pr-guard.yml` |

# 2 Backend: The /search Orchestrator

The search endpoint resolves the question *"what emergency services are near these coordinates, and what are their phone numbers?"* It is implemented in `backend/services/search_service.py` as a four-phase asynchronous pipeline.

## 2.1 Phase 1 — Parallel Geocode and Overpass

Two operations begin simultaneously via `asyncio.gather`: a reverse-geocode against Nominatim and an Overpass query against OpenStreetMap.

```
geo_task = asyncio.create_task(_safe_geocode(lat, lon))
osm_task = asyncio.create_task(_safe_overpass(lat, lon))
geo, osm_contacts = await asyncio.gather(geo_task, osm_task)
```

Listing 1: search_service.py — Phase 1 (lines 127–129)

Wall-clock latency is therefore $\max(t_{\text{geocode}}, t_{\text{overpass}})$ rather than the sum. Both are wrapped in `_safe_*` helpers that catch every exception and return a valid fallback shape; failures never propagate.

### 2.1.1 The Overpass Query Layer

`backend/services/overpass_service.py` builds an Overpass QL query that selects nine categories within a configurable radius (default 5,000 m). The query is structured as a union over tag filters:

```
[out:json][timeout:25];
(
  nwr["amenity"="hospital"](around:5000, LAT, LON);
  nwr["amenity"="clinic"] ["emergency"="yes"](around:5000, LAT, LON);
  nwr["amenity"="police"](around:5000, LAT, LON);
  nwr["amenity"="fire_station"](around:5000, LAT, LON);
  nwr["amenity"="car_repair"](around:5000, LAT, LON);
  nwr["shop"="car_repair"](around:5000, LAT, LON);
  nwr["shop"="tyres"](around:5000, LAT, LON);
  nwr["shop"="car"](around:5000, LAT, LON);
  nwr["emergency"="ambulance_station"](around:5000, LAT, LON);
);
out center;
```

Listing 2: Overpass QL — categories targeted (overpass_service.py:85–119)

If the first response yields fewer than three results, the radius doubles to 10,000 m and the query repeats. This handles sparse regions without paying for the wider radius on dense urban queries.

### 2.1.2 Multi-Mirror Retry Strategy

A single Overpass endpoint frequently rate-limits or 502s. The fetcher rotates across three independent mirrors with exponential backoff:

| Attempt | Endpoint and Backoff |
|---|---|
| 1 | `https://overpass-api.de/api/interpreter` (no delay) |
| 2 | `https://overpass.kumi.systems/api/interpreter` (2 s) |
| 3 | `https://overpass.openstreetmap.fr/api/interpreter` (4 s) |

Each request has a 30-second timeout. If all three mirrors fail, the function returns `[]` rather than raising; downstream phases continue with whatever data exists.

### 2.1.3 Element Parsing and Distance

Each Overpass element is normalised by `parse_element` (overpass_service.py:122–170):

1. Extract `lat`, `lon` (from the element or its `center` field for ways).
2. Compute haversine distance from the user, expressed in kilometres rounded to two decimals.
3. Map the OSM tag to one of nine internal categories: `hospital`, `police`, `ambulance`, `fire`, `towing`, `repair`, `tyre`, `showroom`.
4. Drop results with no name *and* no phone (reduces "Unnamed Building" noise).
5. Parse the `opening_hours` tag into a boolean `isOpen` via `services/hours_parser.py` (supports `24/7`, day ranges, time ranges, and holiday tags).

The haversine implementation is unrolled in pure Python (no GIS dependency):

```
def haversine(lat1, lon1, lat2, lon2):
    R = 6371.0  # Earth radius (km)
    dlat = math.radians(lat2 - lat1)
    dlon = math.radians(lon2 - lon1)
    a = (math.sin(dlat / 2) ** 2 +
         math.cos(math.radians(lat1)) *
         math.cos(math.radians(lat2)) *
         math.sin(dlon / 2) ** 2)
    return 2 * R * math.asin(math.sqrt(a))
```

Listing 3: overpass_service.py:66–78

### 2.1.4 Proximity Deduplication

OSM frequently lists the same facility as both a node (the entrance) and a way (the building polygon). `_dedupe_smart` (overpass_service.py:185–210) keeps results that are either more than 50 m apart *or* have substantially different names; otherwise the earlier-seen entry wins.

## 2.2 Phase 2 — Conditional Google Places Fallback

Phase 2 fires only if Phase 1 produced fewer than three contacts with a dialable phone number. This conserves quota: in dense OSM regions (most of Europe, India, and East Asia), Google is never called.

```
phoned_osm = [c for c in osm_contacts if c.get("phone")]
google_contacts: list[dict] = []
Parallel Google Nearby Search
    google_contacts = await _safe_google(lat, lon, geo.get("country_code"))
```

Listing 4: search_service.py — Phase 2 trigger (lines 135–138)

`services/googleplaces_service.py` issues four parallel Nearby Search calls (hospital, police, car_repair, fire_station) each capped at five results. Per-result Place Details lookups add the `formatted_phone_number` field. The reverse-geocoded country code from Phase 1 is passed as a region hint to improve relevance.

### 2.2.1 Multi-Key Rotation

The `GOOGLE_PLACES_API_KEY` environment variable is parsed as a comma-separated list of API keys. Round-robin rotation distributes load across multiple billing accounts:

```python
def _load_keys() -> list[str]:
    multi = os.getenv("GOOGLE_PLACES_API_KEY", "")
    keys = [k.strip() for k in multi.split(",") if k.strip()]
    if not keys:
        single = os.getenv("GOOGLE_PLACES_API_KEY", "")
        keys = [single] if single else []
    return keys
```

Listing 5: googleplaces_service.py:30–36

If no key is configured, Google is silently disabled — the search still returns whatever Overpass produced.

## 2.3 Phase 3 — Merge and Deduplicate

The two contact lists are merged and deduplicated by phone digits first, then by lowercased name:

```python
def deduplicate(contacts: list[dict]) -> list[dict]:
    out, seen_names = [], set()
    for c in contacts:
        if not isinstance(c, dict): continue
        name_key = (c.get("name") or "").lower().strip()
        if not name_key or name_key in seen_names: continue
        if any(phones_match(c.get("phone"), e.get("phone")) for e in out):
            continue
        seen_names.add(name_key)
        out.append(c)
    return out
```

Listing 6: search_service.py:42–64 — deduplication

`phones_match` (phone_utils.py) compares the last ten digits after `phonenumbers`-normalisation. This handles cases where OSM and Google disagree on transliteration ("GS Custom" vs. "gs sustom") but the phone number is canonical. The merged list is then sorted by distance.

## 2.4 Phase 4 — Phone Enrichment

For the top six closest results without a phone number, the system performs a Find Place from Text lookup against Google Places, biased to a 500-metre radius. On miss, a fallback Nearby Search within 100 m provides a second chance. Each enrichment costs at most one Place Details billable call, capped at six per search to bound worst-case spend.

## 2.5 Response Shape

```json
{
  "contacts": [
    {
      "id": "node/8127364521",
      "name": "Apollo Hospitals, Bannerghatta",
      "category": "hospital",
      "phone": "080-26793000",
      "lat": 12.8932, "lon": 77.5972,
      "distance": 1.42,
      "source": "OpenStreetMap",
      "isOpen": true
    }
  ],
  "source": "OpenStreetMap + Google Places",
  "landmark": "Bannerghatta Road, BTM Layout, Bengaluru",
```

```
  "country_code": "IN",
  "count": 14
}
```

Listing 7: Sample /search response

# 3 Backend: Caching

`backend/services/cache.py` implements an async-safe in-memory TTL cache with LRU eviction. Three module singletons are configured for the dominant lookups:

| Cache | TTL | Max entries | Key strategy |
|---|---|---|---|
| `overpass_cache` | 1 hour | 200 | `round(lat,4)_round(lon,4)_r5000` |
| `google_cache` | 1 hour | 200 | coordinate key + region code |
| `geocode_cache` | 24 hours | 500 | coordinate key only |

## 3.1 Key Granularity

Coordinates are rounded to four decimal places before keying. At the equator one degree of latitude is approximately 111 km, so $10^{-4}$ degrees $\approx$ 11 metres. Two users standing 10 m apart resolve to the same cache key and share results — semantically correct for "what is near me" queries, and trivially exploitable for warm-cache benchmarks (a repeated query from the same demo location returns in tens of milliseconds).

## 3.2 Eviction and Concurrency

- **LRU on capacity.** When `len(cache) >= max_entries`, the entry with the oldest timestamp is removed.
- **Opportunistic expiry.** Each `set` call scans for and removes expired entries before inserting.
- **Async lock.** Each cache instance holds an `asyncio.Lock`; concurrent `get`/`set` operations are serialised per cache.

## 3.3 Negative-Result Policy

Overpass results with zero phone numbers are not cached. This is deliberate: a phoneless cache hit would short-circuit Phase 2 (Google enrichment), permanently denying the user phone-equipped contacts they could otherwise get. The cost is recomputation for sparse regions, accepted as a quality-over-latency trade.

# 4 Backend: Rate Limiting

`services/rate_limiter.py` implements a per-IP token bucket. Each client receives a bucket of capacity 10 that refills at the configured rate; requests consume one token. When the bucket is empty, the request receives `429 Too Many Requests` with a `Retry-After` header in seconds.

| Endpoint | Rate | Burst | Notes |
|---|---|---|---|
| `GET /search` | 30 req/min | 10 | Per-IP |
| `POST /triage` | 20 req/min | 10 | Per-IP, lower because AI calls are costlier |

### 4.1 Render Reverse-Proxy Awareness

Render terminates TLS at its edge and forwards requests with an `X-Forwarded-For` header. The default ASGI `request.client.host` reports the Render edge IP, which would collapse all users into a single bucket. `get_client_ip` (rate_limiter.py:70) prefers the first entry of `X-Forwarded-For` when present:

```
xff = request.headers.get("x-forwarded-for", "")
return xff.split(",")[0].strip() if xff else request.client.host
```

Buckets are stored in a plain Python dictionary protected by `asyncio.Lock`. They reset whenever the Render dyno cold-starts (free tier idle timeout: 15 minutes).

## 5 Backend: AI Triage

`backend/services/ai_triage.py` reorders a contact list to match the user's situation. The model is `gemini-2.5-flash`, called via direct REST against `https://generativelanguage.googleapis.com/v1beta/models/` (no SDK dependency — uses the same `httpx` client as the rest of the backend). The free tier allows 15 requests per minute and 1500 requests per day with no billing requirement. Two boolean situational flags are accepted: `injured` and `blocking_traffic`. JSON mode is opt-in via `generationConfig.responseMimeType = "application/json"`.

### 5.1 Prompt Contract

The system prompt mandates a strict JSON response with exactly two top-level keys:

```
{
  "contacts": [<same items as input, reordered>],
  "reason": "<one short sentence explaining the top pick>"
}
```

Listing 8: Required model output schema

Validation checks (ai_triage.py:87–99) that:

1. the response parses as JSON,
2. `contacts` is a list of exactly the same length as the input,
3. every contact carries the same `id` as an input entry,
4. `reason` is a non-empty string.

### 5.2 Rule-Based Fallback

If the model call fails (no API key, network error, malformed response, validation failure, count mismatch), a deterministic priority table takes over:

| **Situation** | **Priority order (descending)** |
|---|---|
| Injured & blocking | ambulance, hospital, police, towing, repair |
| Injured only | ambulance, hospital, police, repair, towing |
| Blocking only | police, towing, ambulance, hospital, repair |
| Neither | repair, tyre, police, towing, hospital, ambulance |

The fallback is the same logic the model is prompted to follow, so the user-visible behaviour is consistent whether or not the model responded.

### 5.3 Independence from Search

Triage is a separate endpoint (`POST /triage`) called only after the user answers the situational questions. The map and contact list render with no AI dependency; if Gemini is entirely unreachable, the list is still useful.

## 6 Backend: Reliability Guarantees

Every external call is wrapped in a `_safe_*` helper that catches all exceptions and returns a valid fallback shape:

| Wrapper | Fallback return |
|---|---|
| `_safe_overpass` | `[]` |
| `_safe_google` | `[]` |
| `_safe_geocode` | `{"landmark": "latř, lonř", "country_code": null}` |
| `deduplicate` | Skips non-dict items rather than raising |
| `triage_service.prioritise` | Rule-based ordering with outer try/except as last resort |

### 6.1 Catastrophic Failure Path

If every upstream is unavailable simultaneously:

```
{
  "contacts": [],
  "source": "none (upstreams unavailable)",
  "landmark": "13.0827, 80.2707",
  "country_code": null,
  "count": 0
}
```

Listing 9: Response when all upstreams fail

The HTTP status is still `200 OK`. The frontend interprets an empty result as a trigger for the offline fallback chain (Section 7).

### 6.2 Middleware Safety Net

`backend/middleware.py` adds three layers around every request:

1. **RequestIDMiddleware.** Stamps every request with a UUID-v4 (or echoes the client's `X-Request-ID`) and exposes it on the response.
2. **RequestLogMiddleware.** Emits one log line per request: `method path status duration_ms request_id[:8]`.
3. **ErrorHandlingMiddleware.** Outermost. Catches anything an inner handler raised, logs the full traceback at `ERROR` with the request ID, and returns `503 Service Unavailable` with the request ID in the JSON body for support reporting.

## 7 Frontend: Offline Architecture

The frontend implements a four-tier fallback chain. Each tier is queried only if the preceding tier has no answer. The order is fixed in `frontend/src/App.jsx` (lines 197–255).

## 7.1 Tier 1 — Live Backend Search

`searchNearby(lat, lon)` in `src/utils/overpass.js` posts to `/search?lat=&lon=` on the configured backend. Timeout: 8 seconds. On success the response is also written to Tier 2 for future offline use.

## 7.2 Tier 2 — localStorage Cache

`src/utils/offlineDB.js` stores responses in `window.localStorage` keyed by a coarsened coordinate grid:

```
const TTL_MS = 24 * 60 * 60 * 1000; // 24 hours

function locationKey(lat, lon) {
  // Round to 2 decimal places (~1.1 km grid)
  return `roadsos_cache_${lat.toFixed(2)}_${lon.toFixed(2)}`;
}
```

Listing 10: offlineDB.js: 24-hour client-side cache

The grid is 1.1 km square at the equator. A user who searched yesterday from any point within the same square gets a cache hit today. Entries older than the TTL are removed on read.

## 7.3 Tier 3 — Bundled Facilities

When network requests fail and no cache entry exists, `src/utils/bundledFacilities.js` computes the nearest entries from a JSON file shipped in the build bundle: `src/data/bundled_facilities.json` (938 entries, 200 countries).

| Region | Records | Notes |
|---|---|---|
| India (deepest) | ∼40 | Tamil Nadu + Delhi/Mumbai/Bengaluru/Hyderabad metros |
| North America | 22 | US trauma centers (Level I + II), Canadian provinces |
| Europe | 38 | EU member trauma units + UK NHS major centres |
| East and Southeast Asia | 26 | Japan, China, Korea, Thailand, Vietnam, Indonesia, Philippines, Singapore, Malaysia |
| Middle East and North Africa | 18 | UAE, Saudi Arabia, Israel, Turkey, Iran, Egypt |
| Sub-Saharan Africa | 32 | At least one per UN-recognised state |
| Latin America & Caribbean | 24 | Brazil/Argentina/Mexico majors plus smaller-nation hospitals |
| Oceania | 8 | Australia, New Zealand, Pacific island states |

Search radius starts at 80 km and expands to 600 km if zero results match. Distance is computed by the same haversine formula used server-side, ensuring frontend and backend agree on ordering.

## 7.4 Tier 4 — Mock Data

If all three tiers above return empty, a hardcoded array of seven Bengaluru-area contacts (App.jsx:41–112) is used as a last visual placeholder. This tier exists so the UI is never blank; in production deployments it would be removed under a build flag.

### 7.5 Service Worker

frontend/public/sw.js is registered through `vite-plugin-pwa` with the `injectManifest` strategy. The Workbox configuration includes:

- **Precache.** The full app shell (HTML, JS, CSS, fonts) is injected via `self.__WB_MANIFEST` at build time. Total: roughly 670 KiB across six entries.
- **Runtime cache for /search.** `NetworkFirst` with an 8-second timeout, 50-entry LRU, 24-hour max age. After the timeout, the SW serves the most recent cached response for that URL.
- **Runtime cache for /offline-pack.** `CacheFirst` with a single entry and seven-day max age. The backend's offline pack is downloaded once and reused.
- **skipWaiting() on install.** Critical for live deployments. Without it, a new build would only take effect after the user closed every open tab.

### 7.6 Country Emergency Numbers

Separate from facility lookup, src/utils/emergencyNumbers.js is a compile-time mapping from ISO-3166 alpha-2 country code to a list of national emergency dial codes. Examples: `IN` → `[108, 100, 101, 112]`, `US` → `[911]`, `GB` → `[999, 112]`, `DE` → `[112, 110]`. The banner at the top of the app renders these in the user's language and is fully offline-capable.

## 8 Frontend: The Real-Time Map

frontend/src/components/RealMap.jsx renders an OpenStreetMap-backed Leaflet map centred on the user's GPS coordinates.

### 8.1 Tile Provider

Tiles are fetched from CartoDB's Dark Matter raster style:

```
https://{s}.basemaps.cartocdn.com/dark_all/{z}/{x}/{y}{r}.png
```

CartoDB serves these tiles free of charge for non-commercial use with attribution. No API key is required. The `{s}` placeholder distributes across the subdomains `a`–`d`. The `{r}` suffix delivers Retina (`@2x`) tiles on high-DPI displays.

### 8.2 Attribution

OSM's Open Database License (ODbL) and CartoDB's terms both require visible attribution on derived works. The Leaflet attribution control renders bottom-right:

```
&copy; <a href="https://www.openstreetmap.org/copyright">OSM</a>
&copy; <a href="https://carto.com/attributions">CARTO</a>
```

### 8.3 User Marker

A custom `L.divIcon` draws three concentric layers:

1. **Halo.** 110 px radial gradient at 33% opacity, providing visual mass.
2. **Pulse ring.** 28 px circle animated via `@keyframes rs-blip` (2.5 s ease-out infinite).
3. **Dot.** 18 px solid circle with a 3 px white border and a coloured drop-shadow.

When the GPS signal is older than the freshness threshold, the colour shifts from green (`#22C55E`) to grey (`#94A3B8`) and the pulse animation continues at reduced opacity.

## 8.4 Service Markers

Up to six contacts are rendered as map pins using their actual lat/lon coordinates from the search response. Each pin combines:

- A coloured circle (red for medical: hospital, ambulance, fire; blue for police; teal for mechanical: towing, repair, tyre, showroom)
- A category emoji
- A floating chip above with the truncated facility name and distance in kilometres

The map is configured non-interactively in the hero section: `dragging`, `scrollWheelZoom`, `doubleClickZoom`, `touchZoom`, `boxZoom`, and `keyboard` are all disabled. This preserves the SOS button as the primary interaction. A separate full-screen mode (not yet shipped) would enable interaction.

## 8.5 Re-Centering

A child component, `MapRecenter`, hooks into `useMap()` and calls `map.setView()` with a 0.6-second animation whenever the location prop changes. This makes the demo location picker (Bengaluru → London → Tokyo) glide smoothly across the globe.

# 9 Frontend: Internationalization

`frontend/src/i18n/` contains 43 translation bundles registered through `i18next`.

## 9.1 Language Coverage

### 9.1.1 Indian Languages (all 22 Schedule-VIII languages plus English)

Assamese, Bengali, Bodo, Dogri, English, Gujarati, Hindi, Kannada, Kashmiri, Konkani, Maithili, Malayalam, Manipuri, Marathi, Nepali, Odia, Punjabi, Sanskrit, Santali, Sindhi, Tamil, Telugu, Urdu.

### 9.1.2 Global Languages (most-spoken per region)

Spanish, French, Portuguese, German, Italian, Dutch, Polish, Russian, Ukrainian, Romanian, Greek, Turkish, Arabic, Persian, Hebrew, Swahili, Amharic, Indonesian, Malay, Thai, Vietnamese, Chinese, Japanese, Korean, Filipino.

## 9.2 Right-to-Left Support

Six languages set the document direction to RTL on selection (Arabic, Hebrew, Persian, Urdu, Kashmiri, Sindhi). The handler in `src/i18n/index.js`:

```
function applyDir(code) {
  const loc = getLocale(code);
  document.documentElement.lang = loc.bcp47 || code;
  document.documentElement.dir = loc.dir || 'ltr';
}
```

Listing 11: i18n/index.js — applyDir

### 9.3 Selection Logic

On first launch, the user sees a full-screen modal listing all 48 languages, grouped into "India" (22) and "World" (26) sections. There is no automatic pre-selection based on GPS country code; the user picks explicitly. The choice is persisted to `localStorage['roadsos:lang']` and never re-asked.

If the user dismisses the picker without choosing, the language defaults to:

1. The previous session's choice from `localStorage`, if any.
2. `navigator.language` (first two characters), if a translation bundle exists for it.
3. `en`.

### 9.4 Translation Coverage

Each bundle contains approximately 40 translation keys spanning SOS button states, emergency-category labels, location-status messages, action-button labels, crash-alert prompts, and Medical ID form fields. The complete key list is enumerated in `src/i18n/en.json` as the canonical reference.

## 10 Frontend: Crash Detection

`frontend/src/hooks/useLocation.js` fuses two sensor signals to identify a likely crash without requiring dedicated hardware.

### 10.1 Signal 1 — GPS Velocity Collapse

The hook maintains a rolling buffer of GPS samples (each containing speed in m/s and a timestamp). A velocity collapse is registered when sustained samples at $\geq 40$ km/h are followed within 2.5 seconds by a sample at $\leq 5$ km/h (tuned to reject ordinary braking and stop-and-go traffic).

### 10.2 Signal 2 — Accelerometer Spike

A `devicemotion` listener tracks `accelerationIncludingGravity`. A spike is registered when the magnitude exceeds 3.5 G (34.3 $m/s^2$). On iOS 13 and later, motion permission must be granted by an explicit user gesture; the first tap on the SOS area triggers `requestMotionPermission()`.

### 10.3 Confirmation Logic

A crash is reported only if both signals occur within 4 seconds of each other. The two-signal requirement substantially reduces false positives from phone drops (large G spike, no velocity change) and potholes (small velocity change, no sustained G spike).

### 10.4 Cooldown

After firing, the detector enters a 12-second cooldown during which neither signal can re-trigger the alert. This prevents an aftershock or a follow-up impact from doubling notifications.

## 11 Frontend: SOS Dispatch

`frontend/src/utils/sosDispatch.js` translates a button tap into a native messaging app deep link, choosing the dominant medium per country.

## 11.1 WhatsApp-Dominant Countries

For countries where WhatsApp is the de-facto messaging app, the handler issues `window.open('https://wa.me/<ph` `message>')`. The deep link opens WhatsApp (or its web client) pre-filled with:

- The phrase "Emergency. I need help."
- GPS coordinates as decimal degrees.
- A Google Maps link with the same coordinates as the query string.
- The reverse-geocoded landmark string, if available.

WhatsApp-dominant set includes IN, BR, GB, NG, ZA, MX, ID, ES, IT, AR, CO, CL, PE, EG, SA, AE, MY, TR, and roughly 70 others. Because `wa.me` accepts only one recipient per URL, this path sends to a single contact at a time.

## 11.2 SMS-Dominant Countries

For countries dominated by SMS (US, CA, FR, DE, JP, KR, AU, and most of Northern Europe), the handler issues `window.location.href = 'sms:<contact1>,<contact2>,...?body=<encoded message>'`. The native SMS composer opens with all configured emergency contacts pre-populated as recipients.

## 11.3 Event Telemetry

After dispatch, the page emits a `roadsos:sos-sent CustomEvent` with details about the recipient and the medium chosen. `App.jsx` listens for this event and opens `DispatchScreen.jsx` for the post-send confirmation UI. The same event triggers a fire-and-forget `POST /dispatch` to the backend for telemetry.

# 12 Backend API Reference

## 12.1 `GET /search`

| Parameter | Description |
|---|---|
| `lat` | Float, WGS-84 latitude, range $[-90, 90]$. Required. |
| `lon` | Float, WGS-84 longitude, range $[-180, 180]$. Required. |

Returns a JSON object with the shape given in Section 2.5. Rate limit: 30 req/min/IP.

## 12.2 `POST /triage`

```
{
  "contacts": [<list of contact objects from /search>],
  "injured": true,
  "blocking_traffic": false
}
```

Listing 12: Request body

Returns the same contact list, reordered, plus a `reason` string of one short sentence. Rate limit: 20 req/min/IP.

## 12.3 `POST /dispatch`

Telemetry sink. Logs the dispatch event server-side with request ID and returns `202 Accepted`. No body validation beyond JSON parsing.

### 12.4 GET /health

Returns {"status": "ok", "version": "1.x.x", "timestamp": "2026-05-16T..."}. Used by uptime monitors and by the frontend's `startBackendWarmup()` to wake the Render dyno on app load.

### 12.5 GET /offline-pack

Returns the bundled facility JSON for client-side warm caching. `Cache-Control: public, max-age=604800` (7 days).

## 13 Continuous Integration

Three GitHub Actions workflows are defined in `.github/workflows/`.

### 13.1 Frontend CI (frontend-ci.yml)

- **Triggers.** `push` and `pull_request` against `main` when `frontend/**` is touched.
- **Matrix.** Node.js 20 and 22 in parallel.
- **Steps.** `npm ci -legacy-peer-deps` → `vite build` → `vitest run`.
- **Artifact.** The Node-20 job uploads `frontend/dist` as a build artifact (7-day retention).

### 13.2 Backend CI (backend-tests.yml)

Three jobs run sequentially:

1. **lint.** `ruff check` (rule set: E, W, F, I, UP, B, C4, SIM) followed by `ruff format -check`. Four rules are explicitly ignored: E501 (line length), E402 (intentional `load_dotenv()` ordering), B008 (FastAPI `Depends()`), and SIM108 (ternary readability).
2. **startup-check.** Single-step verification that `python -c "import main"` succeeds against `requirements.txt` with dummy API keys — catches dependency-chain breakage before tests run.
3. **test.** `pytest` on Python 3.11 and 3.12 in parallel. Depends on `lint` and `startup-check` passing.

### 13.3 PR Guard (pr-guard.yml)

Two jobs run on every pull request against `main`:

1. **conflict-check.** Attempts a no-commit, no-fast-forward merge of the PR branch against `origin/main`. Fails if Git reports `CONFLICT`, surfacing the issue before review.
2. **branch-staleness.** Counts commits the PR branch is behind `main` via `git rev-list -count HEAD..origin/main`. If the count exceeds 30, emits a warning recommending a rebase. This prevents the kind of large-scale merge conflict that arose during the merge of `arindam-branch-` into `main`.

### 13.4 Branch Protection (Applied at the Repository Level)

The `main` branch enforces:

- Required status checks: Build & Test (Node 20), Build & Test (Node 22), pytest (Python 3.11), pytest (Python 3.12), Lint (ruff), App startup check, and Merge conflict check.

- Strict mode: branches must be up to date with `main` before merging.
- No force pushes.
- No branch deletions.

### 13.5 Test Coverage

Backend tests are organised by service:

| Test file | LOC | Scope |
|---|---|---|
| `test_search_service.py` | 63 | deduplication logic: invalid dicts, empty names, case-insensitive name match, phone equality |
| `test_ai_triage.py` | 90 | rule-based fallback, JSON schema validation, single- and two-leg priority orderings |
| `test_overpass.py` | 207 | element parsing, haversine accuracy, proximity dedup, category classification |
| `test_overpass_parsing.py` | — | edge-case element shapes (nwr without center, tags missing) |
| `test_rate_limiter.py` | — | token-bucket refill, `X-Forwarded-For` extraction, retry-after header |
| `test_cache.py` | — | TTL expiry, LRU eviction, hit/miss statistics |
| `test_geocoder.py` | — | ISO country-code validation against the regex |
| `test_hours_parser.py` | — | `opening_hours` parsing: `24/7`, day ranges, time ranges, public holidays |
| `test_phone_utils.py` | — | E.164 normalisation, `is_dialable` (3–15 digits), `phones_match` |
| `test_middleware.py` | — | request-ID stamping, log line format, error handler 503 response |

## 14 Deployment

### 14.1 Backend on Render

`render.yaml` at the repository root defines a single web service:

```yaml
services:
  - type: web
    name: roadsos-api
    runtime: python
    rootDir: backend
    plan: free
    buildCommand: pip install -r requirements.txt
    startCommand: uvicorn main:app --host 0.0.0.0 --port $PORT
    envVars:
      - key: GEMINI_API_KEY        # required, free at aistudio.google.com
        sync: false
      - key: GOOGLE_PLACES_API_KEY       # optional, comma-separated for multi-key
        sync: false
      - key: PYTHON_VERSION
        value: 3.11.9
```

Listing 13: render.yaml

Free-tier specifics:

- Dyno sleeps after 15 minutes idle; first request triggers a ~30-second cold start.

- No persistent disk (acceptable; all data is upstream or in client browser).
- TLS terminated at edge; backend sees `X-Forwarded-For` only.

`startBackendWarmup()` in the frontend fires a `GET /health` immediately on app mount, paying the cold-start cost before the user runs their first search.

### 14.2 Frontend on Vercel

`vercel.json`:

```
{
  "version": 2,
  "buildCommand": "cd frontend && npm install && npm run build",
  "outputDirectory": "frontend/dist",
  "installCommand": "echo 'install handled in buildCommand'",
  "framework": null,
  "rewrites": [
    { "source": "/(.*)", "destination": "/index.html" }
  ]
}
```

Listing 14: vercel.json

The single rewrite ensures client-side routes resolve through `index.html` (SPA behaviour). Build environment variables:

- `VITE_API_URL` — e.g. `https://roadsos-pl3k.onrender.com`

### 14.3 Bundle Sizes

| Asset | Raw | Gzip |
|---|---|---|
| `dist/assets/index.js` | 610 KiB | 181 KiB |
| `dist/assets/index.css` | 80 KiB | 18 KiB |
| `dist/sw.js` | 25 KiB | 8 KiB |
| `dist/index.html` | 0.8 KiB | 0.4 KiB |
| Precache total | — | ∼210 KiB |

## 15 Observability and Logging

### 15.1 Request Logging

Every request emits one structured log line:

```
2026-05-16T14:32:08Z [INFO] roadsos.request  GET /search 200 1247ms a8f3e9c2
```

Format: timestamp, level, logger, method, path, status, duration in milliseconds, first eight characters of the request ID.

### 15.2 Service-Level Logs

| Service | Notable log events |
|---|---|
| overpass_service | retry attempts (WARN), cache hits (INFO), per-fetch summary (INFO) |
| googleplaces_service | enrichment count per /search (INFO), key rotation (DEBUG) |
| geocoder | invalid country code from Nominatim (WARN) |
| ai_triage | API call failure with exception type (WARN), JSON validation failure (WARN) |
| rate_limiter | per-IP throttling events (INFO) |

### 15.3 Suppressed Noise

`logging_config.py` silences `httpx`, `httpcore`, and `anyio` below WARN level. These otherwise emit a connection-lifecycle line for every internal HTTP call.

## 16 Repository Layout

```
RoadSOS/
|- backend/
|   |- main.py                  # FastAPI app entry, middleware wiring
|   |- middleware.py            # RequestID, RequestLog, ErrorHandling
|   |- logging_config.py        # log format + library noise suppression
|   |- requirements.txt         # runtime deps (FastAPI, httpx, phonenumbers
   , ...)
|   |- requirements-dev.txt     # adds pytest, pytest-asyncio, ruff
|   |- pytest.ini               # asyncio_mode = auto, testpaths
|   |- ruff.toml                # lint rule set + project ignores
|   |- services/
|   |   |- search_service.py        # /search orchestrator (4 phases)
|   |   |- overpass_service.py      # OSM Overpass client + parser
|   |   |- googleplaces_service.py # Google Places + multi-key rotation
|   |   |- geocoder.py              # Nominatim reverse geocode
|   |   |- ai_triage.py             # Gemini 2.5 Flash + rule fallback
|   |   |- triage_service.py        # /triage router
|   |   |- cache.py                 # async TTL/LRU cache
|   |   |- rate_limiter.py          # per-IP token bucket
|   |   |- phone_utils.py           # phonenumbers wrapper
|   |   |- hours_parser.py          # OSM opening_hours parser
|   |   |- health_service.py        # /health endpoint
|   |   |- dispatch_service.py      # /dispatch telemetry endpoint
|   |   `- offline_service.py       # /offline-pack endpoint
|   `- tests/                   # 10 test files, pytest discoverable
|
|- frontend/
|   |- vite.config.js           # Vite + vite-plugin-pwa configuration
|   |- vitest.config.js         # test runner config
|   |- public/sw.js             # Workbox service worker source
|   `- src/
|       |- App.jsx              # top-level orchestrator + state machine
|       |- main.tsx             # ReactDOM root, i18n init
|       |- final-design.css     # design system: colours, typography, layout
|       |- components/
|       |   |- MapHero.jsx      # hero section (map + dock + SOS)
|       |   |- RealMap.jsx      # Leaflet map + custom markers
|       |   |- SOSButton.jsx    # SOS dispatch with WhatsApp/SMS routing
|       |   |- LanguagePicker.jsx # first-launch 48-language modal
|       |   |- MedicalIdModal.jsx # blood type, allergies, contacts
```

```
|      |  |- CrashAlert.jsx        # post-crash confirmation UI
|      |  |- DispatchScreen.jsx    # post-SOS confirmation UI
|      |  |- TriageModal.jsx       # injured + blocking questions
|      |  |- CountryEmergency.jsx # 108/112/911 banner
|      |  `- ...
|      |- hooks/
|      |  |- useLocation.js        # GPS + motion + crash detection
|      |  `- useNetwork.js         # online/offline status
|      |- utils/
|      |  |- overpass.js           # /search client
|      |  |- googlePlaces.js       # /triage client
|      |  |- offlineDB.js          # localStorage cache
|      |  |- bundledFacilities.js # tier-3 fallback search
|      |  |- emergencyNumbers.js  # country -> dial codes
|      |  |- sosDispatch.js        # WhatsApp/SMS deeplink builder
|      |  |- medicalId.js          # on-device emergency-contact storage
|      |  |- plusCodes.js          # OLC encoder (pure JS, no deps)
|      |  `- backendWarmup.js      # Render cold-start mitigation
|      |- i18n/
|      |  |- index.js              # i18next init + RTL handler
|      |  |- locales.js            # locale metadata (native, English, dir)
|      |  |- countryLanguage.js   # ISO country -> language code
|      |  `- *.json                # 43 translation bundles
|      `- data/
|         `- bundled_facilities.json # 938 entries, 200 countries
|
|- docs/
|  |- ARCHITECTURE.md             # architecture review + ADRs
|  `- TECHNICAL.tex               # this document
|
|- .github/workflows/
|  |- frontend-ci.yml             # build + test on Node 20, 22
|  |- backend-tests.yml           # lint + startup + pytest
|  `- pr-guard.yml                # merge conflict + staleness detection
|
|- render.yaml                    # Render service definition
|- vercel.json                    # Vercel build configuration
|- LICENSE                        # MIT
`- README.md                      # human-facing overview
```

## 17 Key Numbers

| Metric | Value |
|---|---|
| Countries with at least one bundled facility | 200 |
| Bundled facilities total | 938 |
| Languages shipped | 43 |
| Indian Schedule-VIII languages covered | 22 of 22 |
| RTL languages supported | 6 (Arabic, Persian, Hebrew, Urdu, Kashmiri, Sindhi) |
| OSM categories queried in parallel | 9 |
| Overpass mirrors with retry | 3 |
| Overpass timeout per attempt | 30 s |
| Default Overpass radius | 5,000 m (expands to 10,000 m if sparse) |
| Bundled-fallback radius | 80 km (expands to 600 km if sparse) |
| Cache TTL — Overpass / Google / geocode | 1 h / 1 h / 24 h |
| Cache key granularity | ∼11 m (4 decimal places) |
| localStorage cache TTL | 24 h |
| localStorage cache grid | ∼1.1 km |
| Rate limit — /search | 30 req/min/IP |
| Rate limit — /triage | 20 req/min/IP |
| Phone enrichment cap | 6 Place Details calls per /search |
| Crash detection — velocity collapse threshold | sustained ≥40 km/h → ≤5 km/h within 2.5 s |
| Crash detection — accelerometer threshold | ≥3.5 G |
| Crash detection — signal alignment window | 4 s |
| Crash detection — post-alert cooldown | 12 s |
| AI model | `gemini-2.5-flash` (free tier: 15 RPM / 1500 RPD) |
| Frontend bundle (gzip) | 181 KiB JS + 18 KiB CSS |
| Service worker precache entries | 6 (∼210 KiB) |

---

**Document version.** Generated from `main` at commit `f94dc2a`. Re-run after substantive code changes; this is a hand-maintained reference, not auto-generated.